MA385: Tutorial Sheet 2

These exercises are for Tutorial 2 (Week 4). You do not have to submit solutions to these questions. However, similar questions may appear on the assignment, class test, or final exam.

In the following questions, take

$$f(x) = x - \cos(x) \tag{1}$$

It may be useful to know that, if $f(\tau) = 0$ then $\tau \approx 0.7390851332$.

Q1. Show that there is a solution to $f(x) = 0$ in the interval $[0, 1]$.

Q2. Is it possible to use Theorem 1.5.2 to determine that Newton's method, applied to f in (1) will converge for any choice of $x_0 \in [0, 1]$? Can it be used to determine that Newton's method will converge for *any* $x_0 \in \mathbb{R}$?

Q3. Take $x_0 = 1$. Use Newton's method x_1, x_2, and x_3 as estimates for solutions to $x - \cos(x) = 0$. If possible, evaluate the corresponding values of $f(x_k)$, and $|\tau - x_k|$, to convince yourself that the method is converging. Use a computer program to do this, if you like.

Q4. Let's suppose we don't know τ, only that it is located between $x = 0$ and $x = 1$. Taking $x_0 = 1$, give upper bounds for $|\tau - x_1|$, $|\tau - x_2|$, and $|\tau - x_3|$ using the *Newton Error Formula*. How does this compare with the corresponding bounds for the Bisection Method, and with the actual errors?